# Project Documentation

## C64 Keyboard Connector PCB

Project number: 224

Revision: 0

Date: 04.06.2026



# C64 Keyboard Connector PCB Rev. 0

## Module Description

## Introduction

The purpose of this PCB is to connect the C64 Keyboard (Project. No. 223 or 225) to the C64/Ultimate64 Keyboard Connector.



*Figure 1: C64 Keyboard Connector PCB with ribbon cable*

The connector for the ribbon cable is a MicroMatch series connector (red). They are low profile, so the cable does not collide with the side of the case when connected to a C64 long board. The C64 Keyboard has a 2x10p pin header, so the ribbon cable has two different types of connectors attached.

## Dimensions

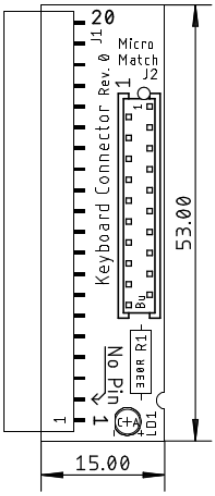



*Figure 2: Dimensions of the PCB*

# Pinouts

## J1 – Connector to mainboard

2.54mm pitch socket, 90°

| Pin | Signal |
|---|---|
| 1 | GND |
| 2 | No pin |
| 3 | /RESTORE |
| 4 | +5V |
| 5 | ROW3 |
| 6 | ROW6 |
| 7 | ROW5 |
| 8 | ROW4 |
| 9 | ROW7 |
| 10 | ROW2 |
| 11 | ROW1 |
| 12 | ROW0 |
| 13 | COL0 |
| 14 | COL6 |
| 15 | COL5 |
| 16 | COL4 |
| 17 | COL3 |
| 18 | COL2 |
| 19 | COL1 |
| 20 | COL7 |

## J2 – Ribbon Cable Connector

MicroMatch Socket, 20 pins

| Pin | Signal |
|---|---|
| 1 | GND |
| 2 | No pin |
| 3 | /RESTORE |
| 4 | +5V |
| 5 | ROW3 |
| 6 | ROW6 |
| 7 | ROW5 |
| 8 | ROW4 |
| 9 | ROW7 |
| 10 | ROW2 |
| 11 | ROW1 |
| 12 | ROW0 |
| 13 | COL0 |
| 14 | COL6 |
| 15 | COL5 |
| 16 | COL4 |
| 17 | COL3 |
| 18 | COL2 |
| 19 | COL1 |
| 20 | COL7 |

# Cable

The cable is a 20p ribbon cable of 45cm with one side attached to a 2x10p IDC socket with strain relief on the keyboard side and a 20 pin MicroMatch connector (male) on the side of this PCB. The cable is shown in the /Cables/pdf/ sub-folder of this repository, drawing number 233-3-01-00 (KB_Cab.pdf).

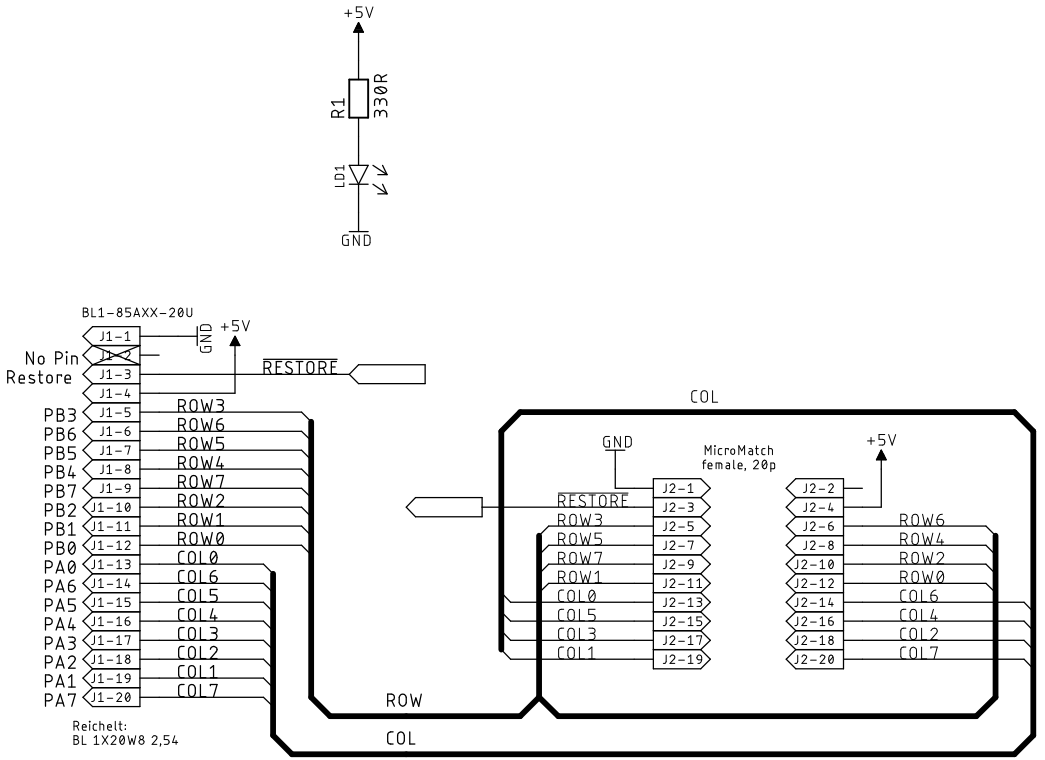
Both connector types are easy to attach, since they are both IDC (insulation-displacement contact), so you only require a vice to compress them, after the ribbon cable is positioned.

The length of 45cm is the longest possible, calculated for a C64 longboard (the keyboard connector is top left on the mainboard). For short boards, 30cm is long enough

# Revision History

## Rev. 0

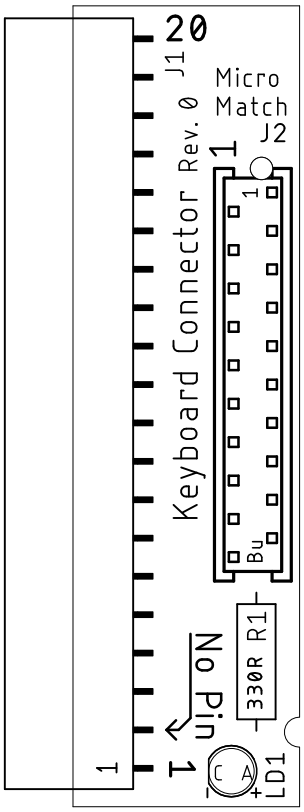
- Working prototype




open source hardware

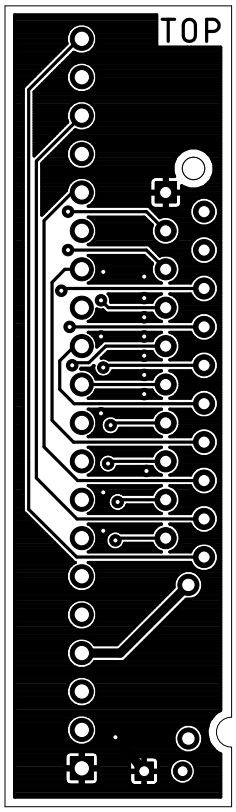
| | | | |
|---|---|---|---|
| Title: | C64 Keyboard Connector PCB | Doc.-No.: | 224-1-01-00 |
| | | Draft.: | Sven Petersen |
| Date: | 20.02.2025 11:05 | Rev.: 0 | Page 1/1 |
| File: | Keyboard_C64_connector_PCB | | |
| | http://github.com/svenpetersen1965 | | A4 |

| Sven Petersen 2025 | Doc.-No.: 224-2-01-00 | |
|---|---|---|
| | Cu: 35µm | Cu-Layers: 2 |
| Keyboard_C64_connector_PCB | | |
| 20.02.2025 19:23 | | Rev.: 0 |
| placement component side | | |

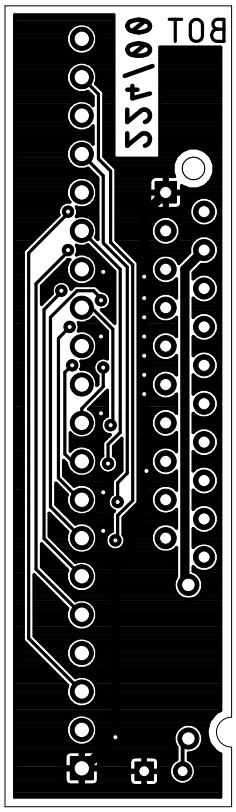


| Sven Petersen 2025 | Doc.-No.: 224-2-01-00 | |
|---|---|---|
| | Cu: 35µm | Cu-Layers: 2 |
| Keyboard_C64_connector_PCB | | |
| 20.02.2025 19:23 | | Rev.: 0 |
| placement solder side | | |

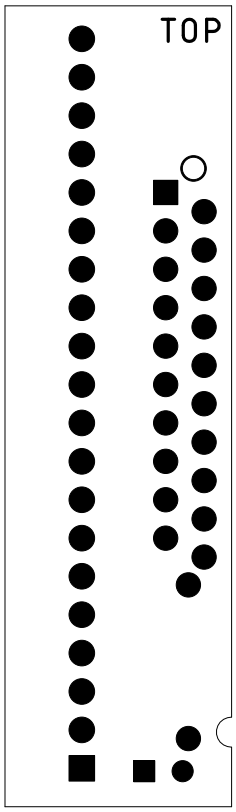
| Sven Petersen 2025 | Doc.-No.: 224-2-01-00 | |
|---|---|---|
| | Cu: 35µm | Cu-Layers: 2 |
| Keyboard_C64_connector_PCB | | |
| 20.02.2025 19:23 | | Rev.: 0 |
| top | [redacted] | |



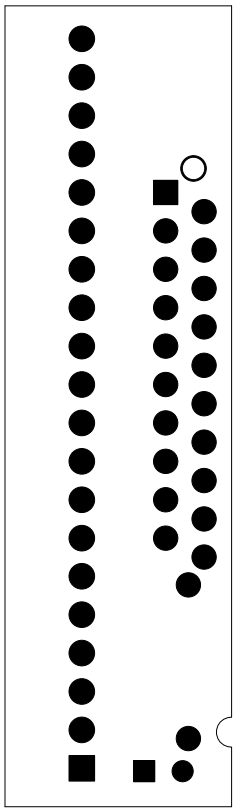
| Sven Petersen 2025 | Doc.-No.: 224-2-01-00 | |
|---|---|---|
| | Cu: 35µm | Cu-Layers: 2 |
| Keyboard_C64_connector_PCB | | |
| 20.02.2025 19:23 | | Rev.: 0 |
| bottom | [redacted] | |



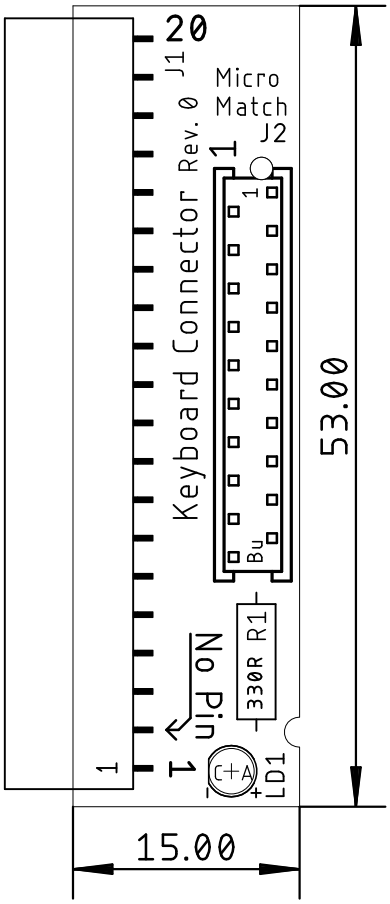
| Sven Petersen 2025 | Doc.-No.: 224-2-01-00 | |
|---|---|---|
| | Cu: 35µm | Cu-Layers: 2 |
| Keyboard_C64_connector_PCB | | |
| 20.02.2025 19:23 | | Rev.: 0 |
| stopmask component side | | |

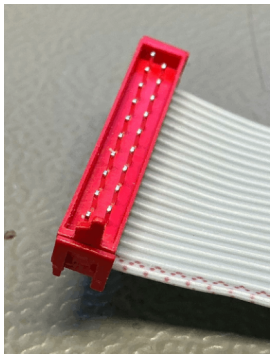


| Sven Petersen 2025 | Doc.-No.: 224-2-01-00 | |
|---|---|---|
| | Cu: 35µm | Cu-Layers: 2 |
| Keyboard_C64_connector_PCB | | |
| 20.02.2025 19:23 | | Rev.: 0 |
| stopmask solder side | | |

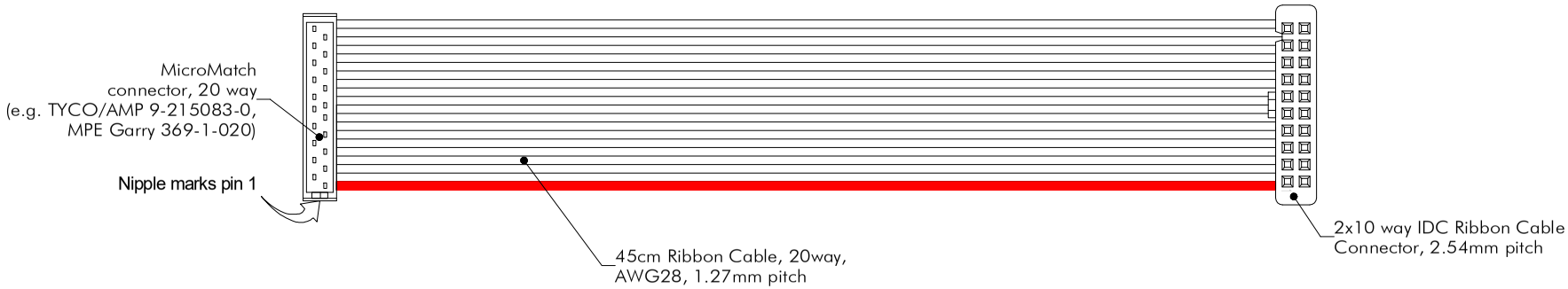


| Sven Petersen 2025 | Doc.-No.: 224-2-01-00 | |
|---|---|---|
| | Cu: 35µm | Cu-Layers: 2 |
| Keyboard_C64_connector_PCB | | |
| 20.02.2025 19:23 | | Rev.: 0 |
| placement component side measures | | |





Detail MicroMatch coonector




This is the length for a long board C64 mainboard.
It varies for a short board or an Ultimate64.

| | | |
|---|---|---|
| open source hardware | Titel: C64 Keyboard Ribbon Cable | Bearb.: Sven Petersen |
| | | Datei: KB_Cab.SKF |
| | Dok.-Nr.: 224-3-01-00 | Rev.: 0 |
| | Datum: 05.06.2026 14:33 | M 1 mm = 1 mm |
| | https://github.com/svenpetersen1965 | |

# C64 Keyboard Connector PCB Rev. 0
## Bill of Material Rev. 0.0

| Pos. | Qty | Value | Footprint | Ref.-No. | Comment |
|---|---|---|---|---|---|
| 1 | 1 | 224-2-01-00 | 2 Layer | PCB Rev. 0 | 2 layer, Cu 35µ, HASL, 53.0mm x 15.0mm, 1.6mm FR4 |
| | 1 | LED, green, 3mm | 3MM | LD1 | Standard LED, 3mm, green |
| | 1 | 330R | R-10 | R1 | Metal film resistor, 1/4W & 10% or better |
| | 1 | BL1-85AXX-20U | BL1-85AXX-20U | J1 | 1x20 pin socket connector, 90°, e.g. MPE Garry, Reichelt BL 1x20W8 2,54 |
| | 1 | 369-1-020 or 2-215079-0 | MICMA20B | J2 | 20 pin MicroMatch, femal, PCB-mount, e.g. MPE Garry, Reichelt MPE 369-1-020 or TE Connectivity 2-215079-0 |
| | 1 | 45cm ribbon cable | | cable | 20 way, AWG28, 1.27mm |
| | 1 | IDC header Micro Match, 20-pin | | cable | Reichelt MPE 369-1-020 or TE Connectivity 9-215083-0 |
| | 1 | 20-pin IDC socket with strain relief, 2.54mm | | cable | e.g. Reichelt PFL 20 |